Reading group notes: mixture-of-experts

Reading group of the Ashbury machine learning lab

1 Background

Most prior methods treat the gating network as fixed and tune the shared expert around it. A practical system has to balance the cost of the router against the quality of the gating network. Despite steady progress, the interaction between the expert and the gating network is still poorly understood. This work studies mixture-of-experts, a method that rethinks the router.

Combining mixture-of-experts with methods that target the expert parallelism is a promising direction. The results suggest that the gating network is the main lever for quality at a fixed budget. The router also makes mixture-of-experts easier to inspect, since its behaviour can be read off per token. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

Recent work has made the expert a central concern, because it decides how far mixture-of-experts models can scale. Despite steady progress, the interaction between the expert parallelism and the top-k routing is still poorly understood. A practical system has to balance the cost of the top-k routing against the quality of the shared expert. Practitioners report that the capacity factor is the first component to fail on long inputs.

2 Methods we read

Unlike a dense model, mixture-of-experts updates the top-k routing only when the expert parallelism changes. The top-k routing is initialised so that mixture-of-experts starts training close to switch routing. We implement the expert as a fused kernel, which avoids writing intermediate results to memory. To keep training stable, mixture-of-experts normalises the load balance before it is combined with the capacity factor.

Unlike switch routing, mixture-of-experts updates the token dropping only when the capacity factor changes. A coefficient controls how strongly the gating network depends on the shared expert; we set it to 0.5 throughout. We implement the expert as a fused kernel, which avoids writing intermediate results to memory. During inference mixture-of-experts caches the router, so each new token costs a constant amount of work.

Mixture-of-experts computes the expert parallelism once per layer and reuses it for every head. The auxiliary loss in mixture-of-experts is parameterised by a small linear layer with 128 hidden units. Unlike switch routing, mixture-of-experts updates the expert only when the capacity factor changes. Gradients through the capacity factor are clipped at a norm of 2.0.

3 Results we discussed

With the top-k routing frozen, mixture-of-experts still recovers most of its improvement on a translation suite. Variance across seeds stays below 0.6 points for every configuration we report. Mixture-of-experts scores 68.7% on a reasoning suite, compared with 93.3% for switch routing. Throughput is measured with a batch size of 16 on a single accelerator.

With the expert frozen, mixture-of-experts still recovers most of its improvement on WebMix perplexity. Variance across seeds stays below 0.7 points for every configuration we report. On a reasoning suite, mixture-of-experts improves over switch routing by 5.7 points. Throughput is measured with a batch size of 16 on a single accelerator.

Ablating the capacity factor lowers the WebMix perplexity score by 5.0 points, the largest drop among all components. With the top-k routing frozen, mixture-of-experts still recovers most of its improvement on a translation suite. Variance across seeds stays below 0.8 points for every configuration we report. Mixture-of-experts scores 51.5% on WebMix perplexity, compared with 83.5% for switch routing.

4 Open questions

We expect the top-k routing to interact with the load balance at larger scales, which we leave to future work. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the expert is the main lever for quality at a fixed budget. A limitation of mixture-of-experts is that the shared expert must be tuned separately for each model size.

The load balance also makes mixture-of-experts easier to inspect, since its behaviour can be read off per token. The results suggest that the expert is the main lever for quality at a fixed budget. We expect the shared expert to interact with the router at larger scales, which we leave to future work. Combining mixture-of-experts with methods that target the shared expert is a promising direction.